

UE23AM441A - Capstone Project Phase - 3

Project Progress Review # 1

Project Title : Explainable Multi-Agent ESG Risk Analysis
System for Indian Companies

Project ID : PW25_MK_01

Project Guide : Dr. Manjula K

Project Team : PES1UG24AM815--Varalakshmi K M
PES1UG24AM807--Malashree H K
PES1UG23AM232--Ranjan C
PES1UG23AM069--Aryan S

Outline

- Abstract and Scope of the Project
- Capstone Project Phase - 2
 - Summary of work
 - Inferences drawn from Literature Survey
- List of Tasks/Modules with Individual Contribution
- Design of overall Architecture
- Demonstration and Testing of the completed modules.

Abstract and Scope

- **Problem** — ESG assessment is increasingly important for investors and regulators, but existing systems often depend on self-reported company disclosures and provide limited transparency or independent verification, creating a risk of **greenwashing**.
- **Objective** — Develop an explainable ESG risk analysis system that combines company disclosures with independent environmental, news and financial data to **validate ESG claims and identify inconsistencies**.
- **Approach** — A sequential **seven-agent pipeline** extracts ESG information from BRSR reports, validates it using external data, performs cross-source analysis, detects potential greenwashing, and generates an explainable ESG assessment.
- **Dataset** — FY 2024-25 BRSR reports of **50 IT companies**, along with CPCB AQI data, ESG-related news and NIFTY 50 historical stock data.

Abstract and Scope

Scope of This Review:

In Scope

- BRSR extraction and ESG metric processing
- AQI-based environmental validation
- ESG news sentiment analysis
- Financial correlation analysis
- Greenwashing detection
- ESG scoring and explainability
- Five Indian IT companies

Not Yet in Scope

- Large-scale coverage of Indian companies
- Real-time ESG dashboard
- Social-media-based validation
- Predictive/time-series ESG forecasting
- Additional governance verification sources

Summary of Work

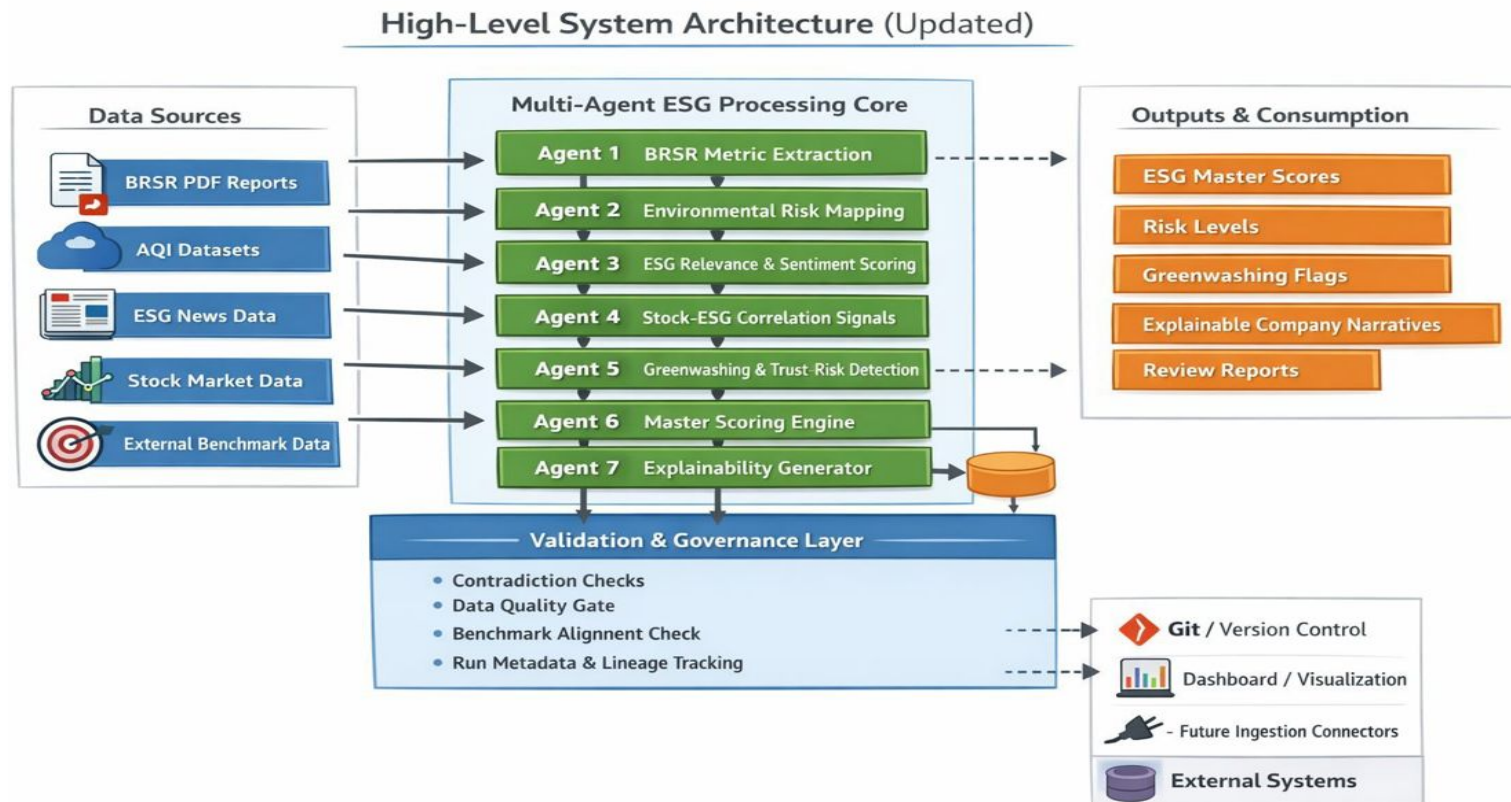
- **Pipeline Built** — Designed and implemented a seven-agent pipeline (extraction, validation, greenwashing detection, scoring, and explainability) covering BRSR extraction, AQI/news/stock cross-validation, and human-readable ESG scoring.
- **Scope Delivered** — End-to-end system validated on 50 major Indian IT companies (TCS, Infosys, Wipro, HCL Technologies, Tech Mahindra) using FY 2024-25 BRSR data, producing 7 structured, auditable output datasets with full data lineage tracking.
- **Greenwashing Detection** — A statistically adaptive three-way cross-check (BRSR claim vs. AQI signal vs. news sentiment) flagged a high-severity contradiction at TCS — a 94% claimed renewable energy figure against a 2.0/10 measured environmental score.
- **Results Achieved** — Classification accuracy improved from 82% to 90% after cross-validation and bias correction. Infosys ranked highest (7.83/10, Low-Medium risk); Tech Mahindra ranked lowest (6.57/10). External benchmark rank correlation of 0.70.
- **Panel Feedback Incorporated** — All seven agents integrated into a single scoring flow, score computation made fully auditable, and the sector-first (IT-only) scope was retained as advised; real-time news ingestion remains in progress.

Inferences from Literature Survey

- **Survey Scope** – 20 peer-reviewed papers and technical reports reviewed across four themes – India-specific ESG/BRSR frameworks, ESG scoring methodologies, multi-agent systems in finance, and explainable AI in finance.
- **Self-Reported ESG Data is Unreliable** – ESG rating-agency correlation averages only 0.54 (vs. 0.99 for credit ratings), motivating an independent, cross-validated scoring approach rather than relying on a single disclosure source.
- **BRSR Compliance is Often Minimal** – Only 17.7% of companies made explicit Net Zero commitments and most comply minimally with disclosure norms, reinforcing the need for a dedicated greenwashing-detection agent.
- **Composite Scores Hide the Real Picture** – Underlying sub-metrics carry more signal than a single aggregate score, and peer-normalization can distort rankings – supporting separate E/S/G pillars scored against absolute BRSR thresholds instead of one opaque number.
- **Multi-Agent Decomposition Outperforms Single-Agent Analysis** – Configurable, specialized agent groups improve accuracy, efficiency, and adaptability for complex financial analysis – validating the project's seven-agent, loosely-coupled pipeline design.
- **Explainability Should Be Local, Not Global** – Finance-domain XAI research favors company-specific, evidence-backed justifications over model-wide explanations, shaping Agent 7 to generate individual reasoning per company.

Architecture

1.Design of overall Architecture



Modules

1.Modules

Agent / Module	Technique Used	Main Function
Frontend	HTML5 + CSS3 + JavaScript	Provides web dashboard and user interface
Backend	Python + FastAPI	Provides backend services and REST APIs
Pipeline Orchestration	<code>demo.py</code> + Subprocess + Multithreading	Executes and coordinates Agent 1–7 pipeline
Agent 1 – BRSR Metric Extraction	PDF extraction + keyword/rule-based matching	Extracts ESG metrics from company BRSR reports
Agent 2 – Environmental Risk Mapping	Location matching + AQI threshold rules	Assesses environmental risk based on company locations and AQI
Agent 3 – ESG News Sentiment Analysis	ESG keyword rules + VADER sentiment analysis	Filters ESG-related news and determines sentiment

Modules

2.Modules

Agent / Module	Technique Used	Main Function
Agent 4 – Stock–ESG Correlation	Statistical analysis + scoring rules	Analyzes stock performance and its relationship with ESG indicators
Agent 5 – Greenwashing & Trust Risk Detection	Rule-based claim/evidence comparison	Identifies potential inconsistencies and greenwashing risks
Agent 6 – Master ESG Scoring	Weighted/rule-based score aggregation	Combines outputs from different agents to generate the overall ESG score
Agent 7 – Explainability Generator	Rule/template-based explanation	Converts ESG scores and risk factors into understandable company-level explanations
Utility Layer	PDF extraction + data validation + data lineage	Supports document processing and data quality
Data Storage	CSV + JSON	Stores agent outputs and final reports

List of Tasks/Modules

1. List of tasks/Modules to be elaborated in discussion with the guide.

1. Data collection and preparation
2. BRSR PDF download, validation, and extraction
3. Environmental and location-based risk analysis
4. ESG news filtering and sentiment analysis
5. Stock market performance and ESG correlation analysis
6. Greenwashing and contradiction detection
7. Master ESG score calculation and quality assessment
8. Explainable AI report generation
9. Cross-validation and data-quality reporting
10. Testing, validation, and performance evaluation

Technologies Used: Python, Pandas, FastAPI, HTML, CSS, JavaScript, CSV/JSON, BRSR reports, AQI datasets, ESG news data, and stock-market datasets.

List of Tasks/Modules

BRSR Data Collection and Preparation

1. Data Collection & Data Preparation

BRSR reports for the configured 50-company cohort were collected from official company, NSE, and BSE sources. Reports were checked for company identity, reporting year, PDF validity, and duplicate files.

2. Data Input

Validated FY 2024-25 BRSR PDFs were used as the primary input. Each report contains environmental, social, and governance disclosures.

3. Data Pre-processing

Text was extracted from PDFs and mapped to standardized ESG metrics. Missing, undisclosed, invalid, and low-confidence values were flagged for review.

4. Data Visualization

Extracted metrics are presented through CSV reports and the web dashboard using company-level ESG summaries and completeness indicators.

5. Data Interpretation

The extracted information is used to evaluate energy consumption, emissions, water usage, employee welfare, governance, and other ESG indicators.

6. Storage

Source PDFs are stored in `brsr-pdfs`. Extracted metrics and data-quality reports are stored in `processed-data` and `demo-output`.

List of Tasks/Modules

Multi-Agent ESG Analysis and Scoring

1. Data Collection & Data Preparation

BRSR metrics, AQI and location data, ESG news articles, and historical stock-market data were collected and aligned using company codes and reporting dates.

2. Data Input

The seven agents receive validated CSV datasets and extracted BRSR metrics as input for environmental, social, governance, sentiment, market, and quality analysis.

3. Data Pre-processing

News articles are filtered for ESG relevance, numerical values are normalized, missing records are identified, and company-level datasets are prepared for analysis.

4. Data Visualization

The frontend dashboard displays company profiles, ESG scores, risk levels, news sentiment, environmental risk, stock correlation, and data-quality results.

5. Data Interpretation

Agent 6 calculates the master ESG score, while Agent 7 generates explanations. Cross-validation and greenwashing checks identify contradictions and review requirements.

6. Storage

Final outputs are stored as CSV and JSON files, including `esg\_master\_scores.csv`, `greenwashing\_detection.csv`, `multi\_agent\_explanations.csv`, and `run\_metadata.json`.

Demonstration and Testing of the Modules Completed

Demonstrated Modules

1. BRSR Extraction: Extracted ESG metrics from company reports.
2. Environmental Risk Analysis: Calculated location-based AQI and environmental risk.
3. News Sentiment Analysis: Classified ESG-related news as positive or negative.
4. Stock Correlation: Compared ESG scores with stock-market performance.
5. Greenwashing Detection: Identified suspicious or contradictory ESG claims.
6. Master ESG Scoring: Generated company-level Environmental, Social, Governance, and overall ESG scores.
7. Explainable AI: Generated explanations for scores, risks, and review flags.
8. Dashboard: Displayed companies, scores, risk levels, sentiment, and data-quality results.

Results Obtained

- 50 company score records
- 900 extracted BRSR metric records
- 677 ESG news sentiment records
- 91 stock-analysis records
- 83 environmental/location-risk records
- Average master ESG score: 6.17/10
- Average BRSR completeness: 40.22%
- 4 companies passed the quality gate
- 46 companies received low-quality warnings
- Cross-validation identified 2 high-severity and 6 medium-severity contradiction

Demonstration and Testing of the Modules Completed

Testing of Completed Modules

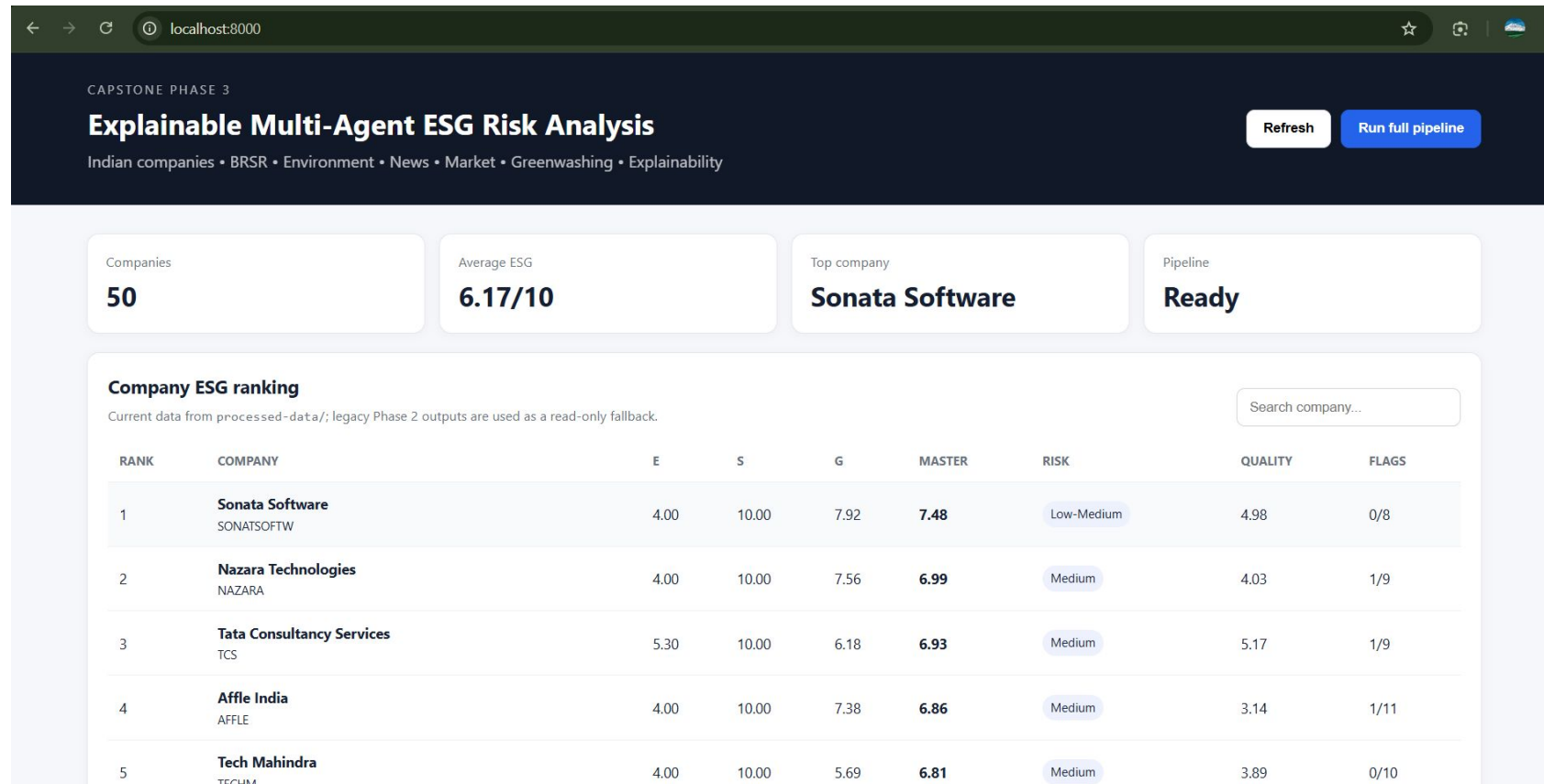
Testing Performed

1. Verified that the configuration contains ****50 companies****.
2. Checked BRSR PDFs for valid file format, company identity, reporting period, and duplicates.
3. Verified that the seven agents execute in the correct sequence.
4. Tested missing, undisclosed, and incomplete BRSR values.
5. Checked that ESG scores and risk levels are generated correctly.
6. Tested news filtering and sentiment classification.
7. Tested contradiction and greenwashing detection.
8. Verified CSV and JSON output generation.
9. Tested backend API endpoints and dashboard data loading.
10. Performed cross-validation and data-quality checks.

Testing Outcome

The pipeline generated the expected output files for the 50-company configuration. Companies with incomplete or unavailable source data were retained as review records instead of being silently removed. The quality-gate process correctly classified records as either ****Pass**** or ****Low-Quality Warning****.

Demonstration and Testing of the Modules Completed



Demonstration and Testing of the Modules Completed

Explainability

Sonata Software Low-Medium

Master ESG score **7.48/10** • data quality 4.98/10

Environmental 4.00/10



Social 10.00/10



Governance 7.92/10



Strengths

Positive media coverage and reputation | Strong board independence (100%)

Weaknesses

Poor air quality at some office locations | Low data quality confidence for scoring inputs

Recommendations

Implement air quality improvement measures at affected locations | Improve data completeness and resolve review-flagged metrics before final decisions | Investigate benchmark rank deviation and document reasons for score differences | Focus on continuous improvement in identified weak areas

Contradictions: 0 **Review flags:** 8

[Master scores CSV](#) · [Greenwashing CSV](#) · [Explainability CSV](#)

Demonstration and Testing of the Modules Completed

BRSR download status

45 latest valid/skipped records • 5 pending

COMPANY	STATUS	REASON
TCS	PASS	company + FY2024-25 + BRSR validated
INFY	PASS	company + FY2024-25 + BRSR validated
WIPRO	PASS	company + FY2024-25 + BRSR validated
HCLTECH	PASS	company + FY2024-25 + BRSR validated
TECHM	PASS	company + FY2024-25 + BRSR validated
LTIM	PASS	company + FY2024-25 + BRSR validated

Demonstration and Testing of the Modules Completed

```
PS C:\Users\mahes\Downloads\capstone-phase3-complete-implementation> .\venv\Scripts\python.exe agents\agent6_master_score.py

AGENT 6: MASTER ESG SCORER
Processing: Tata Consultancy Services, Infosys, Wipro, HCL Technologies, Tech Mahindra, LTIMindtree, Mphasis, Persistent Systems, Coforge, L&T Technology Services, Tata Elxsi, Oracle Financial Services Software, KPIT Technologies, Cyient, Birlasoft, Zensar Technologies, Tata Technologies, Happiest Minds Technologies, Sonata Software, Hexaware Technologies...

=====
CALCULATING MASTER ESG SCORES
=====

🏆 Tata Consultancy Services (TCS)
  Environmental Score: 5.30/10
  Social Score: 10.00/10
  Governance Score: 6.18/10
  Consistency Score: 7.00/10
  Contradictions: 1 (High)
  Quality Score: 5.17/10 (Pass)
  ★ MASTER ESG SCORE: 6.93/10
  Risk Level: Medium

🏆 Infosys (INFY)
  Environmental Score: 6.00/10
  Social Score: 9.00/10
  Governance Score: 4.80/10
  Consistency Score: 10.00/10
  Contradictions: 0 (None)
  Quality Score: 3.61/10 (Low-Quality Warning)
  ★ MASTER ESG SCORE: 6.80/10
  Risk Level: Medium
```

References

- [1] A. Berg, K. Fabisik, and Z. Sautner, “Chasing ESG performance: How methodologies shape ESG scores,” *Journal of Financial Economics*, 2025.
- [2] S. Riedl and M. Smeets, “Evaluation of environmental, social, and governance risks: A critical literature review,” *Sustainability*, vol. 17, no. 18, 2024.
- [3] J. Escrig-Olmedo et al., “The use of ESG scores in academic literature: A systematic literature review,” *Business Strategy and the Environment*, 2023.
- [4] L. Dorfleitner, R. Halbritter, and M. Nguyen, “Deconstructing ESG scores: Investing at the category score level,” *Journal of Asset Management*, 2024.
- [5] H. Liu et al., “Creating a systematic ESG scoring system using social network analysis and machine learning,” *arXiv preprint arXiv:2309.05607*, 2023.

Thank You